PDP

```
!pip install pdpbox
```

```
Collecting pdpbox
  Downloading PDPbox-0.2.1.tar.gz (34.0 MB)
ent already satisfied: pandas in /usr/local/lib/python3.7/dist-
packages (from pdpbox) (1.3.5)
Requirement already satisfied: numpy in /usr/local/lib/python3.7/dist-
packages (from pdpbox) (1.21.5)
Requirement already satisfied: scipy in /usr/local/lib/python3.7/dist-
packages (from pdpbox) (1.4.1)
Requirement already satisfied: joblib in
/usr/local/lib/python3.7/dist-packages (from pdpbox) (1.1.0)
Requirement already satisfied: psutil in
/usr/local/lib/python3.7/dist-packages (from pdpbox) (5.4.8)
Collecting matplotlib==3.1.1
  Downloading matplotlib-3.1.1-cp37-cp37m-manylinux1_x86_64.whl (13.1
MB)
ent already satisfied: sklearn in /usr/local/lib/python3.7/dist-
packages (from pdpbox) (0.0)
Requirement already satisfied: python-dateutil>=2.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib==3.1.1-
>pdpbox) (2.8.2)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.7/dist-packages (from matplotlib==3.1.1-
>pdpbox) (0.11.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!
=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from
matplotlib==3.1.1->pdpbox) (3.0.7)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib==3.1.1-
>pdpbox) (1.4.0)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from kiwisolver>=1.0.1-
>matplotlib==3.1.1->pdpbox) (3.10.0.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.1-
>matplotlib==3.1.1->pdpbox) (1.15.0)
Requirement already satisfied: pytz>=2017.3 in
/usr/local/lib/python3.7/dist-packages (from pandas->pdpbox) (2018.9)
Requirement already satisfied: scikit-learn in
/usr/local/lib/python3.7/dist-packages (from sklearn->pdpbox) (1.0.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from scikit-learn->sklearn-
>pdpbox) (3.1.0)
Building wheels for collected packages: pdpbox
  Building wheel for pdpbox (setup.py) ... e=PDPbox-0.2.1-py3-none-
any.whl size=35758224
sha256=df829e0766ecd9d65850e41a5e514fafe56bb89165735bd7e078c9823312691
```

```
d
  Stored in directory:
/root/.cache/pip/wheels/f4/d0/1a/b80035625c53131f52906a6fc4dd690d8efd2
bf8af6a4015eb
Successfully built pdpbox
Installing collected packages: matplotlib, pdpbox
  Attempting uninstall: matplotlib
    Found existing installation: matplotlib 3.2.2
    Uninstalling matplotlib-3.2.2:
      Successfully uninstalled matplotlib-3.2.2
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
albumentations 0.1.12 requires imgaug<0.2.7,>=0.2.5, but you have
imgaug 0.2.9 which is incompatible.
Successfully installed matplotlib-3.1.1 pdpbox-0.2.1
```

{"pip_warning":{"packages":["matplotlib","mpl_toolkits"]}}

XLRD

```
!pip install xlrd==1.2.0
```

```
Collecting xlrd==1.2.0
  Downloading xlrd-1.2.0-py2.py3-none-any.whl (103 kB)
pting uninstall: xlrd
    Found existing installation: xlrd 1.1.0
    Uninstalling xlrd-1.1.0:
      Successfully uninstalled xlrd-1.1.0
Successfully installed xlrd-1.2.0
```

LIME

```
!pip install lime
```

```
Collecting lime
  Downloading lime-0.2.0.1.tar.gz (275 kB)
ent already satisfied: matplotlib in /usr/local/lib/python3.7/dist-
packages (from lime) (3.1.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.7/dist-
packages (from lime) (1.21.5)
Requirement already satisfied: scipy in /usr/local/lib/python3.7/dist-
packages (from lime) (1.4.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.7/dist-
packages (from lime) (4.63.0)
Requirement already satisfied: scikit-learn>=0.18 in
/usr/local/lib/python3.7/dist-packages (from lime) (1.0.2)
Requirement already satisfied: scikit-image>=0.12 in
/usr/local/lib/python3.7/dist-packages (from lime) (0.18.3)
Requirement already satisfied: PyWavelets>=1.1.1 in
/usr/local/lib/python3.7/dist-packages (from scikit-image>=0.12->lime)
(1.3.0)
Requirement already satisfied: pillow!=7.1.0,!=7.1.1,>=4.3.0 in
```

```
/usr/local/lib/python3.7/dist-packages (from scikit-image>=0.12->lime)
(7.1.2)
Requirement already satisfied: imageio>=2.3.0 in
/usr/local/lib/python3.7/dist-packages (from scikit-image>=0.12->lime)
(2.4.1)
Requirement already satisfied: tifffile>=2019.7.26 in
/usr/local/lib/python3.7/dist-packages (from scikit-image>=0.12->lime)
(2021.11.2)
Requirement already satisfied: networkx>=2.0 in
/usr/local/lib/python3.7/dist-packages (from scikit-image>=0.12->lime)
(2.6.3)
Requirement already satisfied: python-dateutil>=2.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib->lime) (2.8.2)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.7/dist-packages (from matplotlib->lime)
(0.11.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!
=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from
matplotlib->lime) (3.0.7)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib->lime) (1.4.0)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from kiwisolver>=1.0.1-
>matplotlib->lime) (3.10.0.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.1-
>matplotlib->lime) (1.15.0)
Requirement already satisfied: joblib>=0.11 in
/usr/local/lib/python3.7/dist-packages (from scikit-learn>=0.18->lime)
(1.1.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from scikit-learn>=0.18->lime)
(3.1.0)
Building wheels for collected packages: lime
  Building wheel for lime (setup.py) ... e: filename=lime-0.2.0.1-py3-
none-any.whl size=283857
sha256=5628d019b81f0e23c132dc4c697792fdaf6006804469806a4d604063b09cf43
a
  Stored in directory:
/root/.cache/pip/wheels/ca/cb/e5/ac701e12d365a08917bf4c6171c0961bc880a
8181359c66aa7
Successfully built lime
Installing collected packages: lime
Successfully installed lime-0.2.0.1
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.ensemble import RandomForestClassifier

data = pd.read_excel('/content/drive/MyDrive/Colab Notebooks/extra assignment/default of credit card clients.xlsx',skiprows=1)
data.head()
```

```
   ID  LIMIT_BAL  SEX  EDUCATION  MARRIAGE  AGE  PAY_0  PAY_2  PAY_3
PAY_4  \
0   1      20000    2          2         1   24      2      2     -1
-1
1   2     120000    2          2         2   26     -1      2      0
0
2   3      90000    2          2         2   34      0      0      0
0
3   4      50000    2          2         1   37      0      0      0
0
4   5      50000    1          2         1   57     -1      0     -1
0

   ...  BILL_AMT4  BILL_AMT5  BILL_AMT6  PAY_AMT1  PAY_AMT2  PAY_AMT3
\
0  ...          0          0          0         0       689         0

1  ...       3272       3455       3261         0      1000      1000

2  ...      14331      14948      15549      1518      1500      1000

3  ...      28314      28959      29547      2000      2019      1200

4  ...      20940      19146      19131      2000     36681     10000


   PAY_AMT4  PAY_AMT5  PAY_AMT6  default payment next month
0         0         0         0                           1
1      1000         0      2000                           1
2      1000      1000      5000                           0
3      1100      1069      1000                           0
4      9000       689       679                           0

[5 rows x 25 columns]
```

```
data.shape
```

```
(30000, 25)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 25 columns):
```

```
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   ID                          30000 non-null  int64
 1   LIMIT_BAL                   30000 non-null  int64
 2   SEX                         30000 non-null  int64
 3   EDUCATION                   30000 non-null  int64
 4   MARRIAGE                    30000 non-null  int64
 5   AGE                         30000 non-null  int64
 6   PAY_0                       30000 non-null  int64
 7   PAY_2                       30000 non-null  int64
 8   PAY_3                       30000 non-null  int64
 9   PAY_4                       30000 non-null  int64
 10  PAY_5                       30000 non-null  int64
 11  PAY_6                       30000 non-null  int64
 12  BILL_AMT1                   30000 non-null  int64
 13  BILL_AMT2                   30000 non-null  int64
 14  BILL_AMT3                   30000 non-null  int64
 15  BILL_AMT4                   30000 non-null  int64
 16  BILL_AMT5                   30000 non-null  int64
 17  BILL_AMT6                   30000 non-null  int64
 18  PAY_AMT1                    30000 non-null  int64
 19  PAY_AMT2                    30000 non-null  int64
 20  PAY_AMT3                    30000 non-null  int64
 21  PAY_AMT4                    30000 non-null  int64
 22  PAY_AMT5                    30000 non-null  int64
 23  PAY_AMT6                    30000 non-null  int64
 24  default payment next month  30000 non-null  int64
dtypes: int64(25)
memory usage: 5.7 MB

data.describe()

                 ID       LIMIT_BAL           SEX     EDUCATION
MARRIAGE  \
count  30000.000000    30000.000000  30000.000000  30000.000000
30000.000000
mean   15000.500000   167484.322667      1.603733      1.853133
1.551867
std     8660.398374   129747.661567      0.489129      0.790349
0.521970
min        1.000000    10000.000000      1.000000      0.000000
0.000000
25%     7500.750000    50000.000000      1.000000      1.000000
1.000000
50%    15000.500000   140000.000000      2.000000      2.000000
2.000000
75%    22500.250000   240000.000000      2.000000      2.000000
2.000000
max    30000.000000  1000000.000000      2.000000      6.000000
3.000000
```

```
                AGE         PAY_0         PAY_2         PAY_3
PAY_4  \
count  30000.000000  30000.000000  30000.000000  30000.000000
30000.000000
mean      35.485500     -0.016700     -0.133767     -0.166200     -
0.220667
std        9.217904      1.123802      1.197186      1.196868
1.169139
min       21.000000     -2.000000     -2.000000     -2.000000     -
2.000000
25%       28.000000     -1.000000     -1.000000     -1.000000     -
1.000000
50%       34.000000      0.000000      0.000000      0.000000
0.000000
75%       41.000000      0.000000      0.000000      0.000000
0.000000
max       79.000000      8.000000      8.000000      8.000000
8.000000

       ...      BILL_AMT4      BILL_AMT5      BILL_AMT6       PAY_AMT1
\
count  ...   30000.000000   30000.000000   30000.000000   30000.000000

mean   ...   43262.948967   40311.400967   38871.760400    5663.580500

std    ...   64332.856134   60797.155770   59554.107537   16563.280354

min    ... -170000.000000  -81334.000000 -339603.000000       0.000000

25%    ...    2326.750000    1763.000000    1256.000000    1000.000000

50%    ...   19052.000000   18104.500000   17071.000000    2100.000000

75%    ...   54506.000000   50190.500000   49198.250000    5006.000000

max    ...  891586.000000  927171.000000  961664.000000  873552.000000


           PAY_AMT2      PAY_AMT3       PAY_AMT4       PAY_AMT5  \
count  3.000000e+04   30000.00000   30000.000000   30000.000000
mean   5.921163e+03    5225.68150    4826.076867    4799.387633
std    2.304087e+04   17606.96147   15666.159744   15278.305679
min    0.000000e+00       0.00000       0.000000       0.000000
25%    8.330000e+02     390.00000     296.000000     252.500000
50%    2.009000e+03    1800.00000    1500.000000    1500.000000
75%    5.000000e+03    4505.00000    4013.250000    4031.500000
max    1.684259e+06  896040.00000  621000.000000  426529.000000
```

```
            PAY_AMT6  default payment next month
count   30000.000000                30000.000000
mean     5215.502567                    0.221200
std     17777.465775                    0.415062
min         0.000000                    0.000000
25%       117.750000                    0.000000
50%      1500.000000                    0.000000
75%      4000.000000                    0.000000
max    528666.000000                    1.000000

[8 rows x 25 columns]
```

NO null values

checking the labels in y

```
data['default payment next month'].value_counts
```

```
<bound method IndexOpsMixin.value_counts of 0        1
1        1
2        0
3        0
4        0
        ..
29995    0
29996    0
29997    1
29998    1
29999    1
Name: default payment next month, Length: 30000, dtype: int64>
```

```
# for i in data['default payment next month'].values:
#   if i == 1:
#     print('one')
#   elif i == 0:
#     print('zero')
#   else:
#     print('undefined')
```

```
data = data.drop(['ID'],axis=1)
data.head()
```

```
   LIMIT_BAL  SEX  EDUCATION  MARRIAGE  AGE  PAY_0  PAY_2  PAY_3
PAY_4  \
0      20000    2          2         1   24      2      2     -1     -
1
1     120000    2          2         2   26     -1      2      0
0
2      90000    2          2         2   34      0      0      0
0
3      50000    2          2         1   37      0      0      0
0
```

```
4      50000    1          2         1   57     -1      0     -1
0

   PAY_5  ...  BILL_AMT4  BILL_AMT5  BILL_AMT6  PAY_AMT1  PAY_AMT2
PAY_AMT3  \
0     -2  ...          0          0          0         0       689
0
1      0  ...       3272       3455       3261         0      1000
1000
2      0  ...      14331      14948      15549      1518      1500
1000
3      0  ...      28314      28959      29547      2000      2019
1200
4      0  ...      20940      19146      19131      2000     36681
10000

   PAY_AMT4  PAY_AMT5  PAY_AMT6  default payment next month
0         0         0         0                           1
1      1000         0      2000                           1
2      1000      1000      5000                           0
3      1100      1069      1000                           0
4      9000       689       679                           0

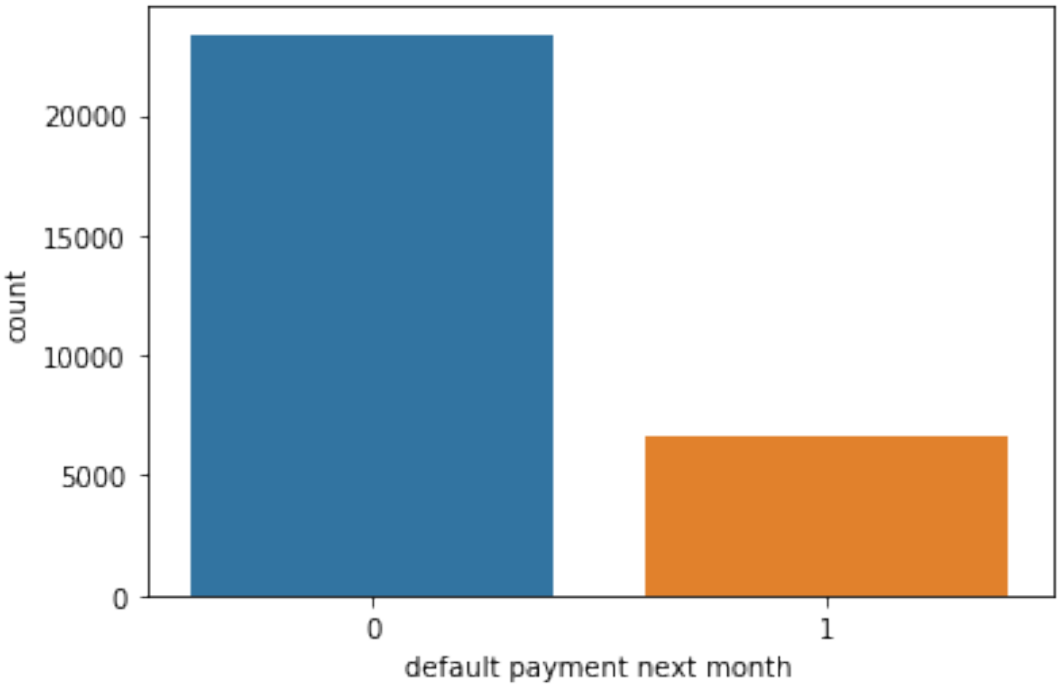
[5 rows x 24 columns]
```

Dataset values

```
sns.countplot(data['default payment next month'])
```

```
/usr/local/lib/python3.7/dist-packages/seaborn/_decorators.py:43:
FutureWarning: Pass the following variable as a keyword arg: x. From
version 0.12, the only valid positional argument will be `data`, and
passing other arguments without an explicit keyword will result in an
error or misinterpretation.
  FutureWarning

<matplotlib.axes._subplots.AxesSubplot at 0x7ffb70ff3690>
```

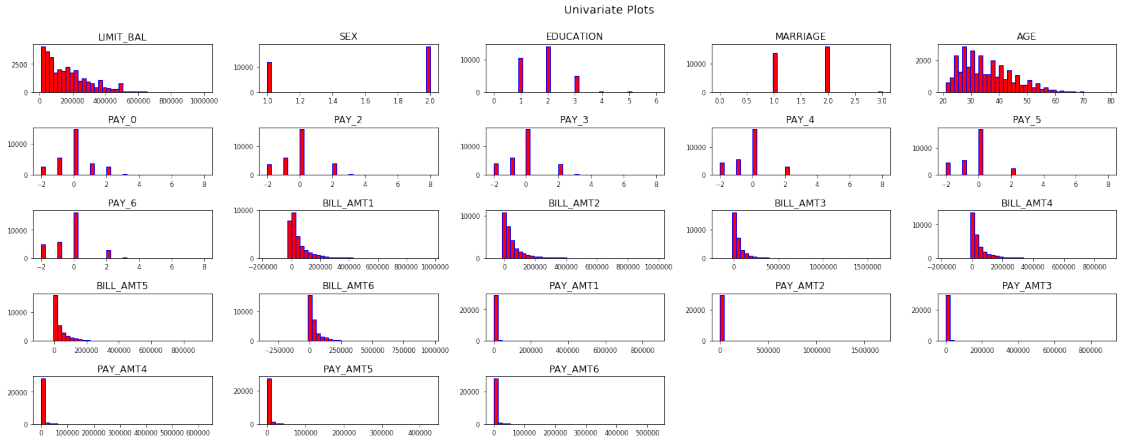


#Quiz EDA and Visualization

```
target = 'default payment next month'
features_list = list(data.columns)
features_list.remove(target)
```

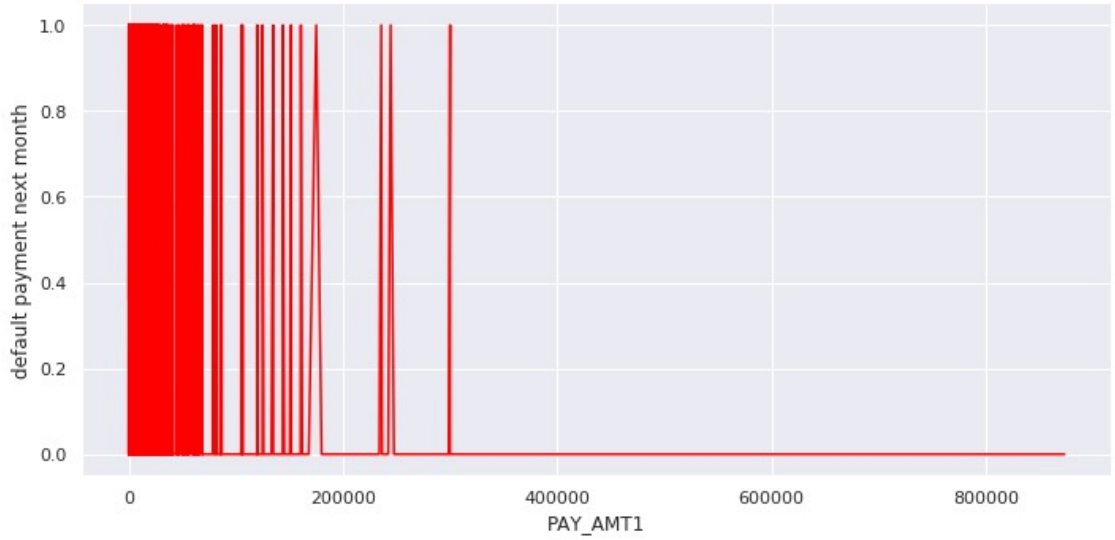
Univariate analysis

```
data[features_list].hist(bins=40, edgecolor='b', linewidth=1.0,
                         xlabelsize=8, ylabelsize=8, grid=False,
                         figsize=(16,6), color='red')
plt.tight_layout(rect=(0, 0, 1.2, 1.2))
plt.suptitle('Univariate Plots', x=0.65, y=1.25, fontsize=14);
```



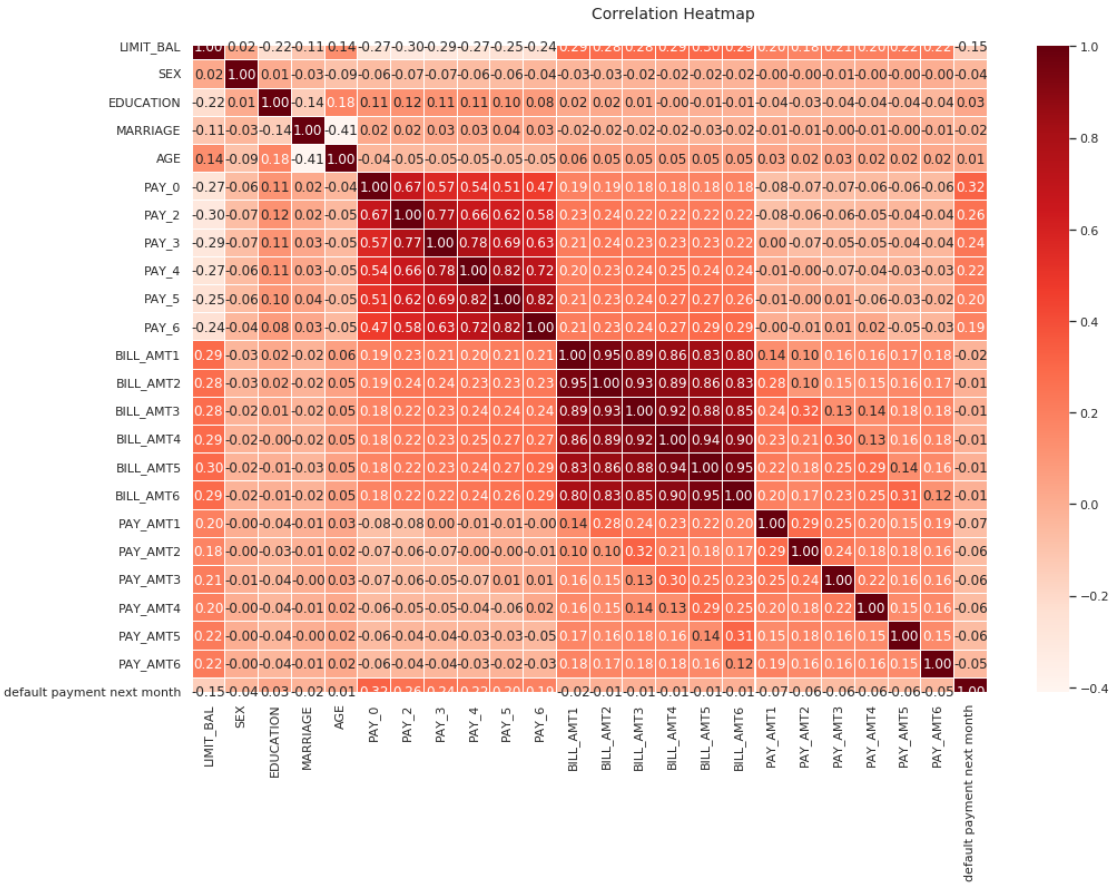
Bivariate analysis

```
sns.set()
sns.relplot(data=data, x='PAY_AMT1', y=target, kind='line', height=5,
aspect=2, color='red');
```



Correlation analysis

```
f, ax = plt.subplots(figsize=(16, 10))
corr = data.corr()
hm = sns.heatmap(round(corr,2), annot=True, ax=ax,
cmap="Reds",fmt='.2f',
            linewidths=.05)
f.subplots_adjust(top=0.93)
t= f.suptitle('Correlation Heatmap', fontsize=14)
```



Feature importance

```
x = data.iloc[:,0:-1]
y = data.iloc[:,-1]
x.shape,y.shape
```

```
((30000, 23), (30000,))
```

```
data.isnull().sum()
```

```
LIMIT_BAL                   0
SEX                         0
EDUCATION                   0
MARRIAGE                    0
AGE                         0
PAY_0                       0
PAY_2                       0
PAY_3                       0
PAY_4                       0
PAY_5                       0
PAY_6                       0
BILL_AMT1                   0
BILL_AMT2                   0
BILL_AMT3                   0
BILL_AMT4                   0
```

```
BILL_AMT5                     0
BILL_AMT6                     0
PAY_AMT1                      0
PAY_AMT2                      0
PAY_AMT3                      0
PAY_AMT4                      0
PAY_AMT5                      0
PAY_AMT6                      0
default payment next month    0
dtype: int64
```

Train test split

```
from sklearn.model_selection import train_test_split as split
x_train,x_test,y_train,y_test = split(x,y,test_size=0.1,stratify=y)
x_train.shape,x_test.shape
```

```
((27000, 23), (3000, 23))
```

Random forest

```
rfc = RandomForestClassifier()

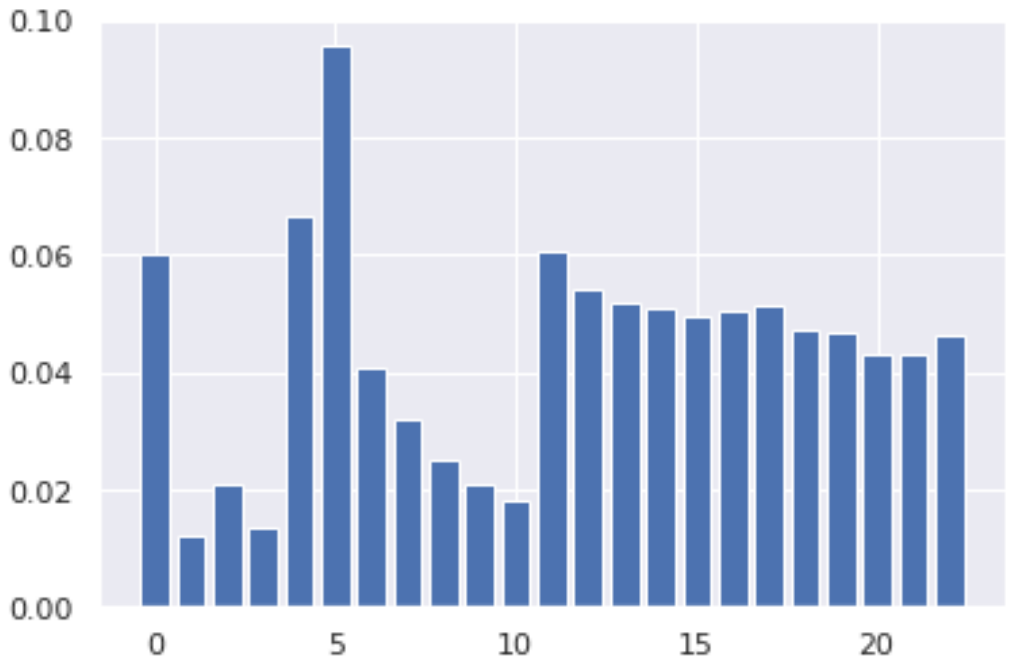
rfc.fit(x_train,y_train)
```

```
RandomForestClassifier()
```

```
importance = rfc.feature_importances_

# for i,v in enumerate(importance):
#     print('Feature: %0d, Score: %.5f' % (i,v))
feature_importance = pd.DataFrame()
feature_importance['variable'] = x_train.columns
feature_importance['importance'] = rfc.feature_importances_
print(feature_importance.sort_values(by='importance',
ascending=False).head(10))
plt.bar([x for x in range(len(importance))], importance)
plt.show()
```

```
     variable  importance
5       PAY_0    0.095577
4         AGE    0.066484
11  BILL_AMT1    0.060404
0   LIMIT_BAL    0.059914
12  BILL_AMT2    0.054258
13  BILL_AMT3    0.052006
17   PAY_AMT1    0.051270
14  BILL_AMT4    0.050951
16  BILL_AMT6    0.050634
15  BILL_AMT5    0.049313
```



(Feature 5) PAY_0 has more importance

```
rfc.score(x_test,y_test)
```

```
0.8186666666666667
```

save model

```
import pickle
pickle.dump(rfc, open("/content/drive/MyDrive/Colab Notebooks/extra assignment/rfc.pkl", 'wb'))
```

LIME

```
import lime
from lime import lime_tabular

interpretor = lime_tabular.LimeTabularExplainer(
    training_data = np.array(x_train),
    feature_names = x_train.columns,
    mode = 'classification'
)

x_test.iloc[5]
```

```
LIMIT_BAL    100000
SEX               1
EDUCATION         1
MARRIAGE          2
AGE              32
PAY_0             0
```

```
PAY_2             0
PAY_3             0
PAY_4             0
PAY_5             0
PAY_6             0
BILL_AMT1     83845
BILL_AMT2     87580
BILL_AMT3     84140
BILL_AMT4     86400
BILL_AMT5     87275
BILL_AMT6     89082
PAY_AMT1       4500
PAY_AMT2       3300
PAY_AMT3       3881
PAY_AMT4       2500
PAY_AMT5       2519
PAY_AMT6       2700
Name: 27171, dtype: int64
```

```
exp = interpretor.explain_instance(
    data_row=x_test.iloc[5], ##new data
    predict_fn=rfc.predict_proba
)

exp.show_in_notebook(show_table=True)
```

```
/usr/local/lib/python3.7/dist-packages/sklearn/base.py:451:
UserWarning: X does not have valid feature names, but
RandomForestClassifier was fitted with feature names
  "X does not have valid feature names, but"

<IPython.core.display.HTML object>
```

Partial Dependence Plots (PD plot)

```
from pdpbox import pdp, get_dataset, info_plots

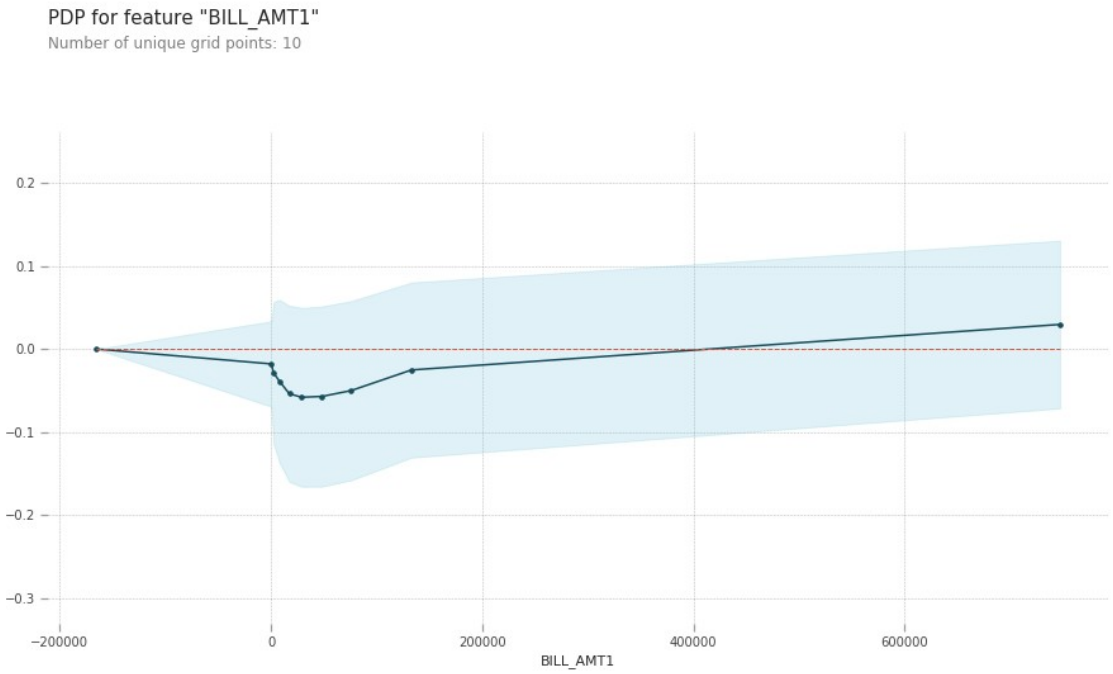
def plot_pdp(model, df, feature, cluster_flag=False, nb_clusters=None,
lines_flag=False):

    # Create the data that we will plot
    pdp_goals = pdp.pdp_isolate(model=model, dataset=df,
model_features=df.columns.tolist(), feature=feature)

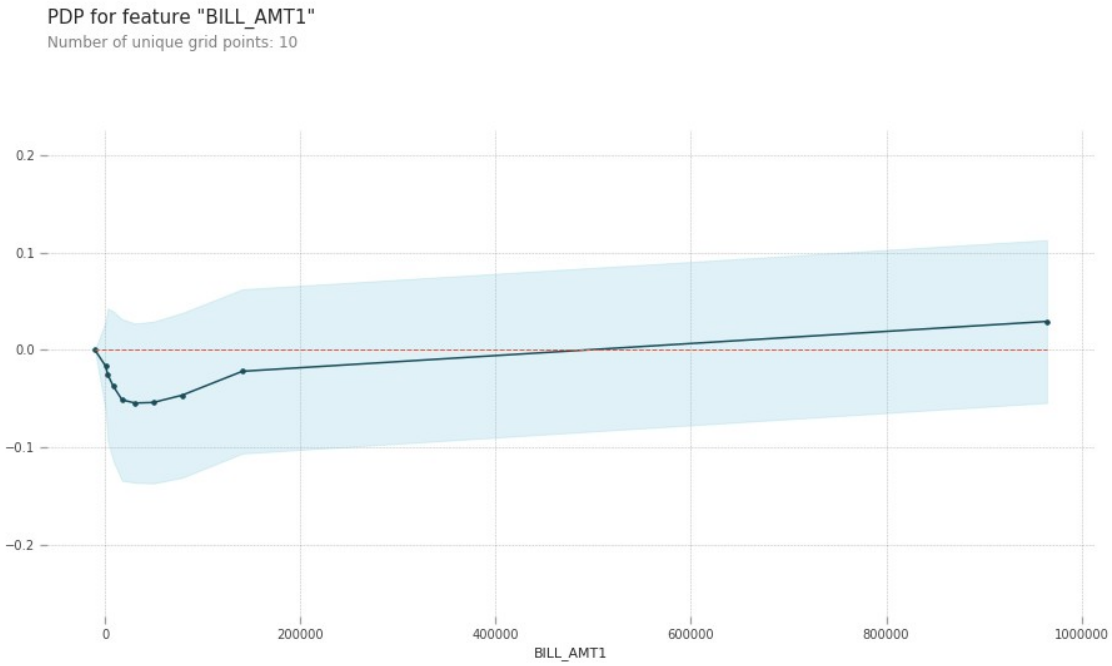
    # plot it
    pdp.pdp_plot(pdp_goals, feature, cluster=cluster_flag,
n_cluster_centers=nb_clusters, plot_lines=lines_flag)
    plt.show()

plot_pdp(rfc, x_train, 'BILL_AMT1')
```

```
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Sans.
```

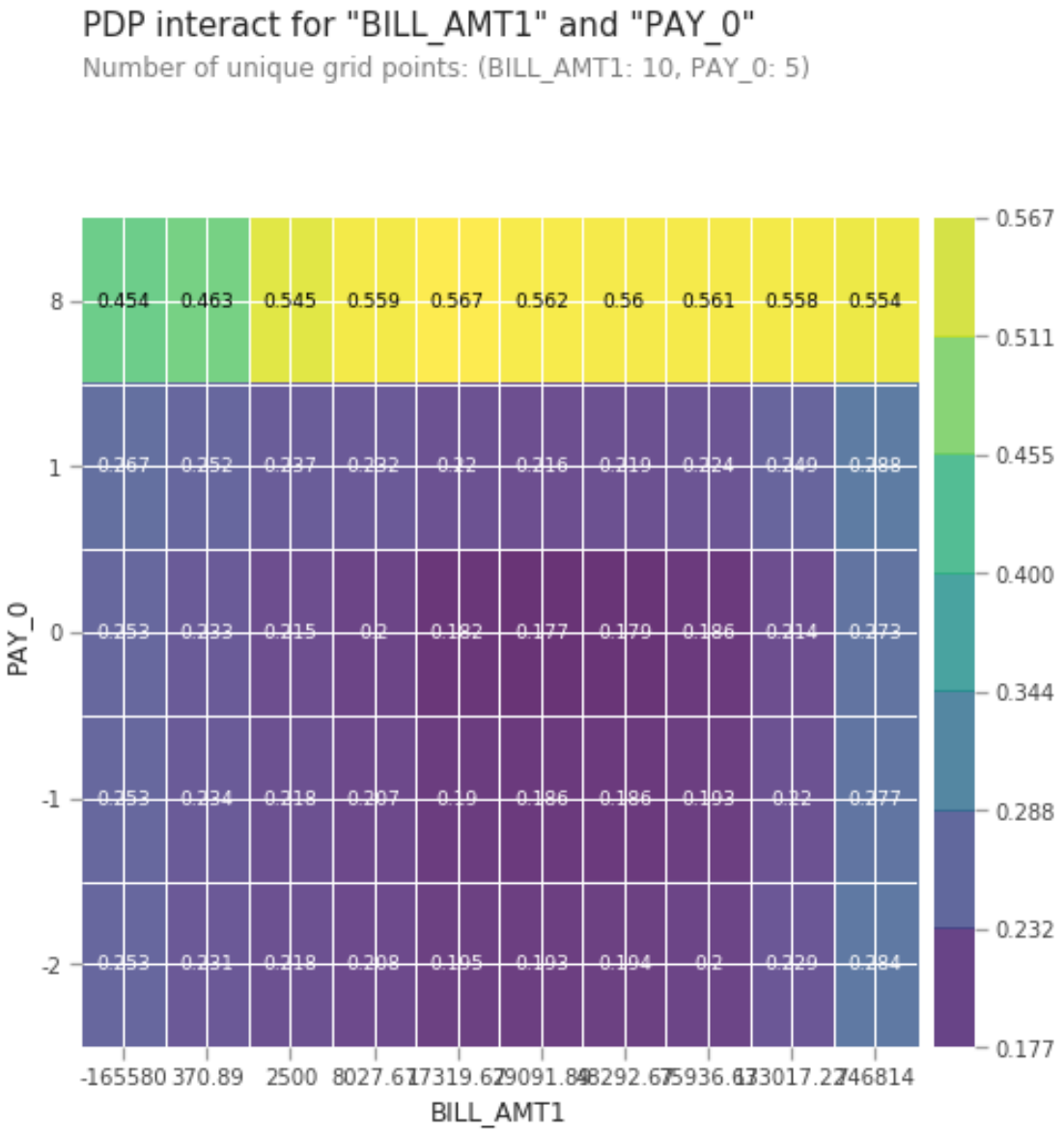



```
plot_pdp(rfc, x_test, 'BILL_AMT1')
```



Bivariate Partial dependancy plot PDP

```
features_to_plot = ['BILL_AMT1', 'PAY_0']
inter1  =  pdp.pdp_interact(model=rfc, dataset=x_train,
model_features=features_list, features=features_to_plot)
pdp.pdp_interact_plot(pdp_interact_out=inter1,
feature_names=features_to_plot, plot_type='grid')
plt.figure(figsize=(24,16))
plt.show()
```



<Figure size 1728x1152 with 0 Axes>

We have used local interpretation along with surrogate models and PDP plots